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Endogenous Growth: Econometric Modelling of the Supply Side

ENDOGENOUS GROWTH AND THE PROCYCLICAL BEHAVIOUR OF PRODUCTIVITY

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In this paper a model with endogenous growth is used to explain the procyclical behaviour of productivity. Periods when the marginal return to goods production is high are associated with low levels of human capital formation and other investment-like activities. In particular, the model successfully explains the surges in productivity exhibited in the UK during wartime and the apparently permanent effect of the level of defence spending on the growth of total factor productivity.

1. Introduction

The procyclical behaviour of productivity is one of the stylised facts which any satisfactory model of economic fluctuations must explain. This fact, along with the relative rigidity of real wages over the business cycle, presents a particular problem to both Keynesian and New Classical paradigms. For with short-run fixity of capital and diminishing marginal returns to labour, both predict anticyclical movements in productivity.

The traditional way to escape from this box is to assume labour hoarding during recessions resulting from hiring and firing costs. However, it is not obvious that this can explain why firms should operate *within* their production frontier during recessions. For provided marginal revenue net of material costs is positive, it must always be profitable for firms to increase production and use up the surplus labour.[1]

An alternative explanation for the procyclical behaviour of productivity is furnished by the Real Business Cycle school. According to this view booms are associated with temporary technology shocks that shift the production frontier outwards and thus with procyclical movements in productivity. However, despite the considerable success of this approach in explaining the

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[1]A more satisfactory version of this story introduces overhead labour as an additional input. There can then be constant, or diminishing, returns to variable labour, but overall productivity can be procyclical. As noted by Hall (1988b), however, this cannot explain the procyclicality of the cost-based total factor productivity residual.

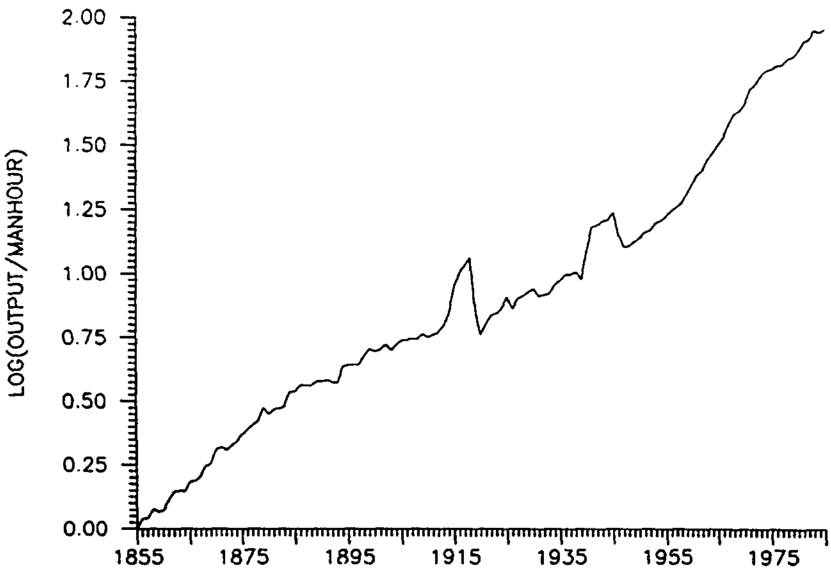

Fig. 1. UK productivity 1855–1985.

temporal behaviour of the main economic aggregates, many economists feel uncomfortable at attributing economic fluctuations entirely to temporary technology shocks. In this regard fig. 1 which plots output per manhour[2] in the United Kingdom since 1855 presents something of a problem, for the most pronounced high-frequency fluctuations in productivity are associated with wars. It seems to be stretching things somewhat to classify the Third Reich as a beneficial shock to technology!

Recently there has been a revival of interest in the mechanics of economic growth and development [Lucas (1988), Romer (1986)]. Traditional growth accounting exercises typically leave a considerable fraction of output growth unexplained by the growth of factor inputs. The new literature attributes this technical progress residual[3] to endogenous human capital formation. I use

[2]Data on hours worked is only available on a continuous basis from 1936 (with a gap over 1939–42), while normal hours is available from 1920. However, Matthews, Feinstein and Odling-Smee (1982) provide observations on actual and normal hours for isolated years prior to this. To construct a proxy for the missing hours worked series, I first interpolated the normal hours series over the pre-1920 period using a polynomial time trend, and then ran a regression of the implied proportion of overtime hours worked on the ratio of weekly earnings to weekly wage rates for the post-1936 data. The estimates were then used to provide an estimate of overtime hours, and hence also hours worked, for the years for which hours worked data was unavailable.

[3]The epithet 'endogenous growth' is somewhat misleading since nobody would argue that the technical progress residual is a genuinely autonomous process that is unaffected by policy e.g. spending on R&D. Rather the issue is the extent to which technical progress is itself affected by the growth process.

this idea to furnish an alternative explanation of procyclical productivity movements. In particular I suggest that during periods when marginal revenue is high (booms), labour is diverted towards the current production of goods and away from human capital formation and other investment-like activities which may not count as output in the official statistics [see also Soderstrom (1977)]. Consequently measured productivity rises. The resulting model rationalises the surges in productivity during wars that are exhibited in fig. 1 as the result of the diversion of manpower from intangible investment towards current production in the face of abnormal levels of government expenditure.

2. An illustrative model

Although I believe the basic idea will extend to other environments, e.g., where economic fluctuations are the result of monetary shocks, I have chosen to illustrate the mechanism with the aid of an optimal stochastic growth model of the sort favoured by Real Business Cycle theorists. This is convenient and has the virtue of clear and coherent microfoundations. The model itself is very similar to that of King and Rebelo (1986).

The representative consumer is endowed with a single unit of labour which is supplied inelastically. However, the effectiveness of that labour depends on the prevailing level of human capital (H_s). Labour and physical capital (K_s) can be combined to produce either physical output (Y_s) or human capital for the next period with the aid of a pair of CES technologies. Physical output may either be consumed by households (C_s), consumed by the government (G_s), or invested in physical capital for the next period. There is complete depreciation of both sorts of capital. Goods market clearing requires that ($s \geqq t$)

$$C_s + G_s + K_{s+1} = U_s[\alpha(M_s K_s)^\rho + (1-\alpha)(N_s H_s)^\rho]^{1/\rho} = Y_s, \tag{1}$$

where the right-hand side of the equality is the output of goods ($\rho < 1$), U_s is a technology shock and M_s (N_s) is the proportion of the economy's physical capital (labour endowment) used in goods production. Human capital is given by the accounting identity ($s \geqq t$)

$$H_{s+1} = V_s\{\alpha[(1-M_s)K_s]^\rho + (1-\alpha)[(1-N_s)H_s]^\rho\}^{1/\rho}, \tag{2}$$

where the right-hand side is the production function for human capital and V_s is another technology shock. Note that for simplicity I have assumed that, with the exception of the shocks, the technologies for goods and human capital are identical.

The representative consumer is infinitely-lived and derives no benefit from government spending. His objective function is given by the intertemporally separable function

$$U(C_t, C_{t+1}, \ldots) = \sum_{s=t}^{\infty} \delta^{s-t} C_s^{\rho}/\rho, \tag{3}$$

where δ is the consumer's discount factor. Note that the index of relative risk aversion, $(1-\rho)$, is constrained to equal the inverse of the elasticity of substitution in production. While this is again special, it facilitates a simple solution.

Government spending is financed by lump-sum taxation and, once more for simplicity, I shall assume that it is the *share* of government expenditure in total physical output, rather than the level, that is exogenously given. Let $\{W_s\}$ denote the associated sequence of the share of private expenditure in total physical output.

Because there are no externalities, distortionary taxes, etc., in this economy, the Pareto optimum can be supported as a competitive equilibrium. The latter can therefore be obtained by solving a social planning problem which maximises the expectation of the objective function (3), conditional on information available at date t, and subject to the laws of motion (1) and (2). The realisation of the vector (U_t, V_t, W_t) is assumed to be known to the planner (and the private sector) at date t, but not earlier. The stochastic sequence $\{U_s, V_s, W_s\}$ is assumed to be serially independent. Necessary first-order conditions ($s \geqq t$) are then

$$N_s = M_s, \tag{4}$$

$$E_t[C_s^{\rho-1}] = \alpha\delta E_t[(U_{s+1}W_{s+1})^{\rho}(C_{s+1}M_{s+1}K_{s+1}/W_{s+1}Y_{s+1})^{\rho-1}], \tag{5}$$

$$E_t[(U_sW_s/V_s)C_s^{\rho-1}] \\ = (1-\alpha)\delta E_t[(U_{s+1}W_{s+1})^{\rho}(C_{s+1}N_{s+1}H_{s+1}/W_{s+1}Y_{s+1})^{\rho-1}], \tag{6}$$

where $E_t[\cdot]$ is the expectation operator conditional on date t information. Serial independence of the shocks ensures that the solution to these equations must be of the form $C_s = \pi W_s Y_s$ and $K_{s+1} = (1-\pi)W_s Y_s$, where π is a constant. The Euler equations (5) and (6) together with (2) then imply

$$(1-\pi)^{1-\rho} = \alpha\delta E_t[(U_{t+1}W_{t+1})^{\rho} M_{t+1}^{\rho-1}], \tag{5'}$$

$$(1-M_t)/M_t = (1-\pi)[(1-\alpha)/\alpha]^{1/(1-\rho)}(V_t/U_tW_t)^{\rho/(1-\rho)}, \tag{6'}$$

where the right-hand side of (5′) is a constant that depends on the probability distribution of the shocks. It follows from (6′) that a decrease in (U_tW_t/V_t), i.e., an increase in government spending, a negative shock to the goods technology or a positive shock to the human capital technology, leads to a shift in resources towards marketable output if and only if the elasticity of substitution in production is less than unity ($\rho<0$). Hence an increase in government spending will also lead to an increase in measured productivity under precisely these conditions.

What do these results imply about the evolution of the two capital stocks? Substitution into the laws of motion (1) and (2) gives:

$$K_{t+1}=\frac{(1-\pi)(\alpha U_tW_t)^{\sigma}[\alpha K_t^{\rho}+(1-\alpha)H_t^{\rho}]^{1/\rho}}{[\alpha^{\sigma}(U_tW_t)^{\rho\sigma}+(1-\pi)(1-\alpha)^{\sigma}V_t^{\rho\sigma}]}, \tag{7}$$

$$H_{t+1}=\frac{(1-\pi)[(1-\alpha)V_t]^{\sigma}[\alpha K_t^{\rho}+(1-\alpha)H_t^{\rho}]^{1/\rho}}{[\alpha^{\sigma}(U_tW_t)^{\rho\sigma}+(1-\pi)(1-\alpha)^{\sigma}V_t^{\rho\sigma}]}, \tag{8}$$

where $\sigma=1/(1-\rho)$ is the elasticity of substitution in production. Consequently we have the following responses to shocks: $\mathrm{d}K_{t+1}/\mathrm{d}U_t$, $\mathrm{d}H_{t+1}/\mathrm{d}V_t$, $\mathrm{d}K_{t+1}/\mathrm{d}W_t>0$; and $\mathrm{d}H_{t+1}/\mathrm{d}U_t$, $\mathrm{d}K_{t+1}/\mathrm{d}V_t$, $\mathrm{d}H_{t+1}/\mathrm{d}W_t\gtrless 0$ as $\rho\lesseqgtr 0$. Eqs. (7) and (8) also imply that these transitory shocks will have permanent effects. While King, Plosser and Rebelo (1988) have already pointed out this result for technology shocks, a similar effect from fiscal policy has to my knowledge not been noted before. Thus the finding that a large part of the variance in output appears to be permanent [e.g. Nelson and Plosser (1982)] does not necessarily say much about the original source of economic fluctuations. Finally, since

$$K_{t+1}/H_{t+1}=[\alpha U_tW_t/(1-\alpha)V_t]^{\sigma}, \tag{9}$$

it follows that $\ln K_t$ and $\ln H_t$, despite being non-stationary, are co-integrated [again see King, Plosser and Rebelo (1988)].

3. Empirical evidence

The distinctive feature of the theory laid out above is not that (the logarithms of) K_t and H_t are co-integrated – a conventional growth model in which H_t is an exogenous process would also predict that – but rather: (i) that H_t is not strictly econometrically exogenous; and (ii) transitory shocks e.g. due to wartime defence spending, have permanent effects on the levels of H_t and K_t, i.e. pure hysteresis. To test these hypotheses[4] we first need to construct proxies for H_t. This I do by cumulating the total factor productivity residuals. Logarithmically differentiating the goods production technology gives

$$\Delta y_t = S_{kt}(\Delta m + k_t) + (1 - S_{kt})(\Delta n_t + \Delta h_t) + \Delta u_t, \tag{10}$$

where lower-case letters denote logarithms, Δ is the difference operator, and S_k is the share of capital in output. Of course one needs to allow for changes in hours and employment, so in the empirical work below Y_t will be identified with output per manhour, and K_t with the capital–labour ratio. H_t should thus be thought of as human capital per head. Unfortunately Δm_t, Δn_t and Δu_t are all unobservable so as a proxy I use

$$\begin{aligned}\Delta h_t^* &= (\Delta y_t - S_{kt}\Delta k_t)/(1 - S_{kt}) \\ &= \Delta h_t + [S_{kt}\Delta m_t + (1 - S_{kt})\Delta n_t + \Delta u_t]/(1 - S_{kt}),\end{aligned} \tag{11}$$

which is then cumulated to give a noisy measure of human capital.

This approach assumes that factors receive their marginal products. However, if there is imperfect competition in product markets, labour receives less than its marginal product and the weight on physical capital in (10) is too big. Hall (1988a) suggests estimating the extent to which price exceeds marginal cost by carrying out an instrumental variable regression of the Solow total factor productivity residual on the capital–labour ratio. Unfortunately this procedure will not work in the presence of endogenous growth because all potential instruments will be correlated with Δm_t and Δn_t. Instead I follow Hall (1988b) in using the share of capital in cost, rather than revenue, as a measure of S_{kt} in constructing an alternative human capital proxy.

Eqs. (8) and (9) suggest estimating a VAR sub-system of the form

$$x_{t+1} = A(L)x_t + B(L)e_t, \tag{12}$$

where $x_t = [\Delta h_t^*, h_t^* - k_t]'$, e_t is any set of proxies for $\{u_t, v_t, w_t\}$ and $A(L)$, $B(L)$ are comformable matrices in lag polynomials. Exogeneity of the human capital formation process then requires that no variable Granger-cause Δh_t^*,

[4] I believe these properties will also hold outside the narrow confines of the model of section 2.

i.e. $A_{12}(L) = B_{1i}(L)$ (for all i) = 0. The hypothesis of hysteresis can be tested by examining whether a permanent shock has a long-run effect on the growth rate of human capital. Letting $A_{ij} = A_{ij}(1)$, etc., the absence of hysteresis requires, for all i,

$$[(1-A_{22})B_{1i}+A_{12}B_{2i}]/[(1-A_{11})(1-A_{22})-A_{12}A_{21}]=0. \tag{13}$$

Before turning to the results, note that the presence of the measurement error in h_t^* will not affect the validity of either of these tests. Under the null hypothesis that h_t is exogenous, Δm_t and Δn_t are both zero in (11). Hence h_t^* will still be strictly econometrically exogenous, provided u_t is also exogenous (which is reasonable). However, the causality test does lack power against a Keynesian-style alternative where procyclical productivity results from demand shocks and labour hoarding. Then true h_t could be exogenous, but h_t^* will now be a poor proxy because labour input is mis-measured, and the difference between h_t and h_t^* is likely to be correlated with lagged variables that fluctuate with the business cycle. The test of hysteresis is robust, however, because even where measurement error does lead to spurious Granger-causality, as in the labour-hoarding case, it should only affect the short-run dynamics and not the estimated long-run effect.

These tests are applied to British data, 1855–1987, drawn from Mitchell (1988), supplemented by official publications for the recent past. Y_t is GDP per man hour and K_t is the ratio of the gross fixed capital stock to the employed civilian labour force. All the tests are carried out for two measures of h_t^*, one employing the share of capital in total revenue, the other employing the share of capital in total cost. For the latter, capital costs are obtained by multiplying the capital stock by a measure of the user cost of capital. This in turn is obtained as the predicted value from a second-order autoregression of the ex post real short-term interest rate, plus a 10 per cent allowance for depreciation. The exogenous driving variable, e_t, is the (logarithm of) one minus the share of output allocated to defence expenditures. Only military expenditure is included because other forms of government spending may be substitutes for consumption. By contrast it seems reasonable to assume that military spending yields no direct utility.

Although not relevant for distinguishing between endogenous and exogenous productivity growth, table 1 reports the results of tests for co-integration between each measure of h_t^* and k_t [see Engle and Granger (1987) for further details of these tests]. The first column gives the estimated slope coefficient from a regression of h_t^* on k_t (expected to be unity) and its reported standard error. The second column reports the co-integrating Durbin–Watson statistic (95 per cent critical value = 0.39). The last column gives the result of an augmented Dickey–Fuller test on the residuals from the co-integrating regression, obtained by estimating a fourth-order autoregres-

Table 1
Tests of co-integration.

	Co-integrating vector	Durbin–Watson	Augmented Dickey–Fuller
Revenue-based residual	1.12 (0.02)	0.24	−0.104 (2.08)
Cost-based residual	1.006 (0.019)	0.27	−0.134 (2.48)

Table 2
Hypothesis tests.

	Granger-causality F(5, 121)	Long-run coefficient
Revenue-based residual	13.91	0.0358 (2.64)
Cost-based residual	9.84	0.0403 (2.91)

sion in the first difference of the residual, augmented by a single lagged level. The table reports the coefficient and t-statistic on this last variable (95 per cent critical value = 3.17).

While the evidence in favour of co-integration is not overwhelming, I nevertheless proceed to the next stage involving the estimation of a second-order VAR in Δh_t^* and the residual from the co-integrating regression, augmented by three lags of the transformed military spending variable. For brevity the full results for each regression are not reported. However, table 2 reports the results of tests of the main hypotheses. The first column reports the results of a Granger-test of the exogeneity of h_t^*. This is rejected at very high significance levels. As noted above, however, while this serves to reject a primitive real business cycle model with exogenous technical progress, it is not inconsistent with a Keynesian-style labour-hoarding model. The second column is more informative in this regard. It gives the long-run effect on the growth rate of human capital of a permanent shock to the military spending variable, together with its asymptotic t-statistic. The estimates imply that a one percentage point increase in the share of output devoted to military spending eventually lowers the growth rate of human capital and output by about 0.04 percentage points, and in both cases this effect is statistically significant. This is not predicted by conventional models, but is consistent with the sort of endogenous growth model laid out above.

References

Engle, R.F. and C.W. Granger, 1987, Co-integration and error correction: Representation, estimation and testing, Econometrica 52, 1241–1269.

Hall, R.E., 1988a, The relation between price and marginal cost in US industry, Journal of Political Economy 96, 921–947.
Hall, R.E., 1988b, Increasing returns: Theory and measurement with industry data, Mimeo. (Stanford University).
King, R.G. and S.T. Rebelo, 1986, Business cycles with endogenous growth, Mimeo. (University of Rochester).
King, R.G., C.I. Plosser and S.T. Rebelo, 1988, Production, growth and business cycles: II. New directions, Journal of Monetary Economics 21, 309–342.
Lucas, R.E., 1988, On the mechanics of economic development, Journal of Monetary Economics 22, 1–42.
Matthews, R.C.O., C.H. Feinstein and J.C. Odling Smee, 1982, British Economic Growth 1856–1973 (Oxford).
Mitchell, B.R., 1988), British historical statistics (Cambridge University Press).
Nelson, C.R. and Plosser, C.I., Trends and random walks in macroeconomic time series: Some evidence and implications, Journal of Monetary Economics 10, 139–162.
Romer, P.M., 1986, Increasing returns and long-run growth, Journal of Political Economy 94, 1002–1037.
Soderstrom, H.T., 1977, Team production and the dynamics of the firm, Scandinavian Journal of Economics 79, 271–288.